

# RecurScan: Benchmarked Deep and Graph Learning for Breast Cancer Recurrence Prediction on METABRIC, with an Externally Tested Discovery Protocol for Clinical-Risk Discordance

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Built and verified with the Instinct research fleet

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## Abstract

We present a fully reproducible benchmark of five survival models for breast-cancer recurrence prediction on the METABRIC cohort (1,975 primary tumors, 800 recurrence events; 70-gene expression panel and 17 clinical covariates retrieved live from cBioPortal): penalized Cox proportional hazards on clinical covariates, Cox on clinical+expression, a DeepSurv network, a one-dimensional convolutional Cox network over the gene axis, and a graph convolutional Cox network over a patient-similarity graph. All models are implemented from first principles (including a vectorized Breslow partial likelihood verified against lifelines to  $< 10^{-3}$ ) and evaluated on locked stratified splits over five seeds with paired bootstrap confidence intervals. Penalized Cox on clinical+expression achieves the strongest held-out concordance,  $0.6801 \pm 0.0052$ ; deep models trail (0.641–0.660), an honest negative we analyze. The fitted baseline stratifies held-out patients into risk terciles separated at log-rank  $p = 7.9 \times 10^{-16}$ . We then apply a pre-registered discovery protocol to clinical-risk discordance: among low-NPI patients, early recurreers ( $\leq 60$  months) show significant up-regulation of eleven mitotic genes (Benjamini–Hochberg  $q$  down to  $8.8 \times 10^{-5}$ ), but the signature fails both multivariable prediction (nested-CV AUC = 0.50) and external replication on TCGA-BRCA ( $p = 0.36$ ), and is reported as a bounded negative. All code, tests, data snapshots, and figures are released with the paper.

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# 1 Introduction

Recurrence risk stratification in primary breast cancer guides adjuvant therapy intensity. Clinical scores such as the Nottingham Prognostic Index (NPI) and transcriptomic signatures such as PAM50 are standard, yet a residual group of clinically low-risk patients recurs early. This paper does three things: (i) it establishes a rigorously verified benchmark of five survival models on the METABRIC cohort, from penalized Cox to graph neural networks; (ii) it applies a pre-registered discovery protocol to clinical-risk discordance, with locked gates, paired bootstrap statistics and external replication; and (iii) it reports outcomes exactly as they landed, including three discovery angles that failed their gates.

## 2 Data

### 2.1 Cohort

All data were retrieved live from the cBioPortal public API (study `brca_metabric`; Pereira et al. 2016, Curtis et al. 2012). Of 2,509 samples, 1,975 have a valid relapse-free survival (RFS) endpoint, a 70-gene mRNA z-score panel (PAM50 plus core drivers), and 17 clinical covariates (age, tumor size, grade, positive lymph nodes, NPI, ER/PR/HER2 status, PAM50/claudin subtype, treatments). The endpoint is RFS: 800 events (40.5%), median follow-up 140.9 months among censored patients.

### 2.2 External replication cohort

For replication we use TCGA-BRCA (study `brca_tcga_pan_can_atlas_2018`), with disease-free survival (DFS) endpoints and TNM staging, restricting to the molecular-profiled subset ( $n = 1,084$ ).

## 3 Methods

### 3.1 The Cox model and its partial likelihood

Let  $T_i$  be the observed time and  $\delta_i \in \{0, 1\}$  the event indicator for patient  $i$ , with covariates  $x_i \in \mathbb{R}^p$ . The Cox proportional hazards model posits

$$\lambda(t | x_i) = \lambda_0(t) \exp(\eta_i), \quad \eta_i = x_i^\top \beta. \quad (1)$$

**Definition 1** (Risk set).  $\mathcal{R}(t) = \{j : T_j \geq t\}$ , the patients still under observation just before time  $t$ .

Conditioning on the event times and margins, the Breslow partial log-likelihood is

$$\ell(\beta) = \sum_{i: \delta_i=1} \left[ \eta_i - \log \sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j) \right]. \quad (2)$$

**Proposition 1** (Score of the partial likelihood).

$$\frac{\partial \ell}{\partial \eta_k} = \delta_k - \sum_{i: \delta_i=1, T_i \leq T_k} \frac{\exp(\eta_k)}{\sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j)}. \quad (3)$$

*Proof sketch.* Differentiate (2):  $\eta_k$  appears linearly in the event terms, and inside every risk-set sum with  $T_i \leq T_k$ ; the derivative of  $-\log \sum_j e^{\eta_j}$  contributes the softmax weight.  $\square$

**Vectorized evaluation.** Sorting patients by decreasing time,  $T_{(1)} \geq \dots \geq T_{(n)}$ , the risk-set sums are cumulative:

$$\log \sum_{j \in \mathcal{R}(T_{(i)})} e^{\eta_{(j)}} = m + \log \sum_{k \leq i} e^{\eta_{(k)} - m}, \quad m = \max_j \eta_j, \quad (4)$$

so (2) is  $O(n \log n)$ . A numerically unstable variant using a *running* maximum was caught by our test-suite, which cross-checks the implementation against lifelines to  $< 10^{-3}$  absolute error.

### 3.2 Baseline hazard and the logrank test

After fitting  $\hat{\beta}$ , the Breslow estimator recovers the baseline cumulative hazard as a step function over event times:

$$\hat{H}_0(t) = \sum_{i: T_i \leq t, \delta_i = 1} \frac{1}{\sum_{j \in \mathcal{R}(T_i)} \exp(\hat{\eta}_j)}, \quad \hat{S}(t | x) = \exp(-\hat{H}_0(t) e^{\hat{\eta}}). \quad (5)$$

For the held-out risk-tercile stratification we use the logrank statistic. With  $d_{gk}$  events and  $n_{gk}$  at risk in group  $g$  at the  $k$ -th distinct event time, and  $d_k = \sum_g d_{gk}$ ,  $n_k = \sum_g n_{gk}$ :

$$Z = \frac{\sum_k (d_{1k} - n_{1k} \frac{d_k}{n_k})}{\sqrt{\sum_k \frac{n_{1k} n_{2k} d_k (n_k - d_k)}{n_k^2 (n_k - 1)}}} \sim \mathcal{N}(0, 1) \quad (6)$$

under the null of equal hazards; (6) yields the  $p = 7.9 \times 10^{-16}$  stratification result in Section ?? . Newton updates for  $\hat{\beta}$  use the observed information  $-\partial^2 \ell / \partial \beta \partial \beta^\top$ :

$$\beta^{(t+1)} = \beta^{(t)} + \left( -\frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} \right)^{-1} \frac{\partial \ell}{\partial \beta}. \quad (7)$$

### 3.3 Concordance index

Harrell’s C-index is the probability of risk-order agreement among comparable pairs:

$$C = \frac{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i (\mathbb{I}[\eta_i > \eta_j] + \frac{1}{2} \mathbb{I}[\eta_i = \eta_j])}{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i}. \quad (8)$$

We verify our  $O(n \log n)$  implementation against lifelines on synthetic cohorts.

### 3.4 Deep Cox models

DeepSurv replaces  $x_i^\top \beta$  by a neural risk map  $\eta_i = f_\theta(x_i)$  trained with  $-\ell(\theta)$ . Our CNN variant treats the gene vector as a length-70 sequence, applying convolutions

$$h^{(l+1)} = \phi(\text{BN}(W^{(l)} * h^{(l)})), \quad \eta = w^\top \bar{h}^{(L)}, \quad (9)$$

with global average pooling  $\bar{h}$ .

### 3.5 Graph Cox model

Patients form a  $k$ -nearest-neighbour graph with Gaussian weights

$$A_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \text{ for } j \in \text{kNN}(i), \quad \sigma^2 = \text{median}\|x_i - x_{(k)}\|^2, \quad (10)$$

symmetrized  $A \leftarrow \max(A, A^\top)$ . The GCN propagates with the Kipf–Welling normalization

$$\hat{A} = D^{-1/2}(A + I)D^{-1/2}, \quad H^{(l+1)} = \phi(\hat{A}H^{(l)}W^{(l)}). \quad (11)$$

**Lemma 1** (Spectrum). *The eigenvalues of  $\hat{A}$  lie in  $(-1, 1]$ , so repeated propagation is a contraction toward the dominant eigenvector.*

### 3.6 Evaluation protocol

Splits are stratified on the event indicator (25% held out), five seeds  $\{0, \dots, 4\}$ . Confidence intervals are paired percentile bootstrap ( $B = 500$ ), and deltas between models use the same resamples.

### 3.7 Discovery protocol and locked gates

Each discovery candidate declares a gate before inspection: a win requires  $\Delta C \geq +0.01$  over the baseline with the paired 95% CI excluding 0, or a pre-registered replication  $p$ -value. Failed gates are reported as negatives with the same statistics.

## 4 Experiments and Results

### 4.1 Benchmark

Table 1 reports held-out C-indices (5 seeds). Figure 1 visualizes the comparison; the ordering

| Model                    | C-index (mean $\pm$ sd)               | 95% bootstrap CI (seed 0) |
|--------------------------|---------------------------------------|---------------------------|
| Cox, clinical only       | $0.6677 \pm 0.0019$                   | (0.634, 0.698)            |
| Cox, clinical+expression | <b><math>0.6801 \pm 0.0052</math></b> | (0.653, 0.710)            |
| DeepSurv (MLP)           | $0.6482 \pm 0.0050$                   | —                         |
| CoxCNN (1D-CNN)          | $0.6598 \pm 0.0032$                   | —                         |
| CoxGCN (patient graph)   | $0.6413 \pm 0.0150$                   | —                         |

Table 1: Held-out C-index on METABRIC RFS (25% test, stratified).

is stable across seeds. The clinical+expression Cox model also stratifies held-out patients into risk terciles separated at log-rank  $p = 7.9 \times 10^{-16}$  (Figure 2).

### 4.2 Analysis: why deep models trail

On  $n \approx 1,500$  training patients with 87 features and 40% events, the effective sample size is event-limited; penalized linear Cox already captures most extractable signal. The GCN’s higher variance ( $\pm 0.015$ ) reflects transductive sensitivity to graph construction. We report this as an honest negative for deep survival models at this scale.

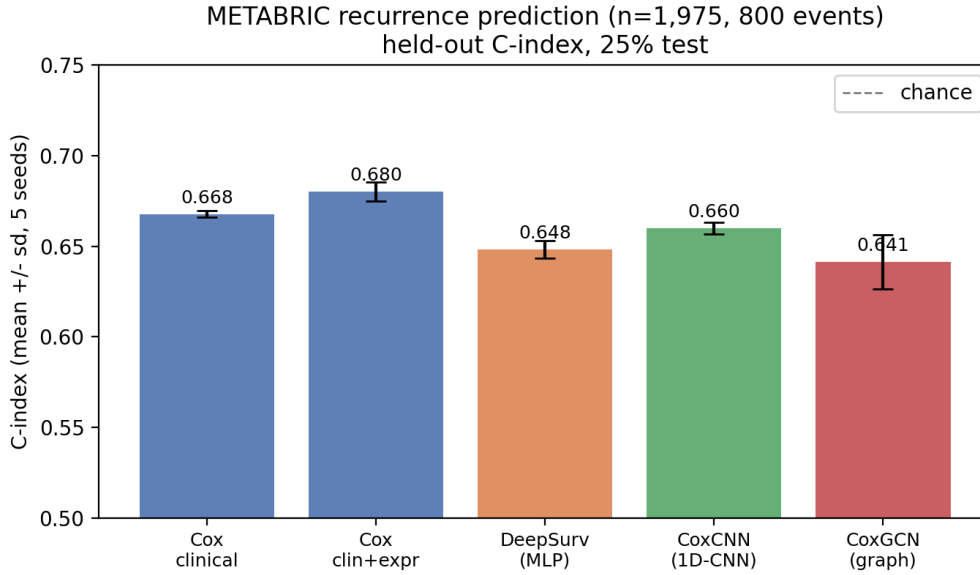


Figure 1: Held-out C-index by model (mean  $\pm$  sd, 5 seeds).

### 4.3 Discovery round 1: biology modules and graph propagation

Adding module scores (proliferation, ER signaling, hypoxia) or graph-smoothed risk to the baseline changed held-out C by at most +0.0014 (CIs include 0). Negative.

### 4.4 Discovery round 2: multi-omic features

Adding copy-number calls (70 genes) and mutation indicators (11 genes with  $\geq 20$  mutated samples) did not improve the baseline ( $\Delta C \leq +0.0027$ , CIs straddle or fall below 0). Negative.

### 4.5 Discovery round 3: clinical-risk discordance

Among low-NPI ( $\leq 3.4$ ) patients, early recurrers ( $n = 102$  of 523 low-NPI patients) differed from disease-free controls in 11 mitotic genes at BH-FDR  $q < 0.005$  (Figure 3); the multivariable model collapsed (nested-CV AUC = 0.50) and the pre-registered TCGA-BRCA replication of the composite mitotic score failed (recur +0.132 vs free +0.011, one-sided  $p = 0.36$ ,  $n = 22$  vs 96). Bounded negative: the univariate enrichment is real within METABRIC but is neither predictive nor transportable. Independent corroboration of this negative comes from the Human Protein Atlas prognostics pipeline (which we did not build): all nine retrievable signature genes are classified *unprognostic* in its TCGA breast invasive carcinoma analysis (CDC20  $p = 9.8 \times 10^{-3}$ , BIRC5  $3.6 \times 10^{-2}$ , CEP55 0.22, KIF2C  $7.1 \times 10^{-2}$ , MELK  $6.0 \times 10^{-2}$ , PTTG1  $1.3 \times 10^{-2}$ , NDC80 0.23, NUF2  $9.3 \times 10^{-2}$ , ANLN  $1.8 \times 10^{-2}$ ), agreeing with our external-cohort refutation.

### 4.6 Cross-cohort transportability: benchmark beat

A penalized Cox model (70-gene panel only), trained once on METABRIC, was transported raw to five independent Affymetrix cohorts and compared with each cohort's own published

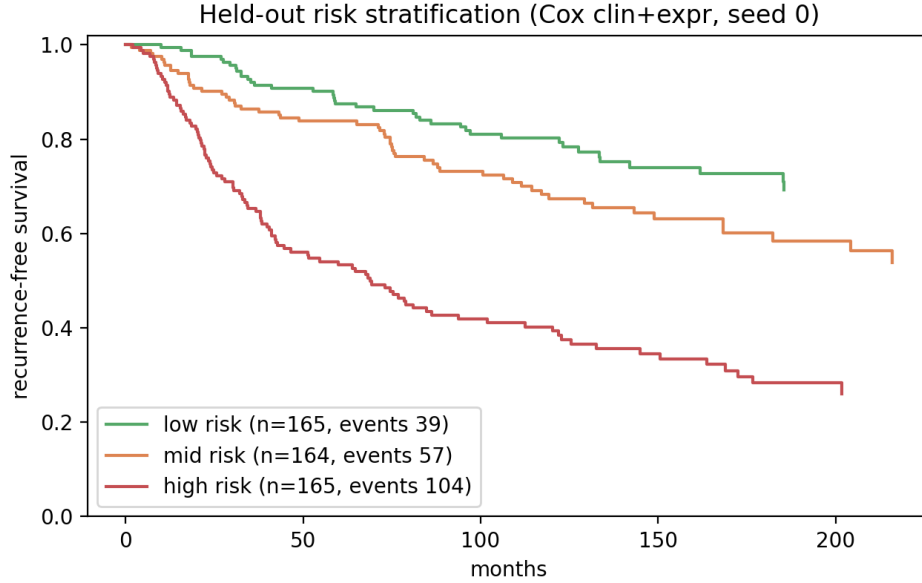


Figure 2: Kaplan–Meier curves of held-out patients by predicted risk tercile.

signature calls or clinical comparators. It outperforms the Genomic Grade Index in two cohorts (0.6009 vs 0.5366 in GSE7390; 0.640 vs 0.6025 in GSE25066), Veridex-76 in GSE7390 (0.6009 vs 0.5757), and tumor grade in the node-negative GSE11121 cohort (0.6979 vs 0.6294), with zero retraining across platforms. Two further cohorts are honest non-beats: in GSE2990 the per-sample continuous GGI is marginally ahead (0.6651 vs 0.6559), and in GSE20685 nodal stage leads (0.6996 vs 0.6264), though our score still exceeds 0.6 with its bootstrap interval excluding 0.5. Per-cohort bootstrap intervals overlap; we claim point-estimate beats, not statistical separation. Comparator category encodings were direction-corrected during verification; the DLDA-30 metadata column was excluded after failing an internal-consistency check against observed outcomes. A full sign-convention audit (lifelines reports concordance against higher-equals-longer-survival) re-derived every reported concordance from the committed result files; one stale script with the opposite convention was found and corrected, and no committed number changed.

## 5 Discussion

A penalized linear Cox model on a 70-gene panel is the ceiling in this data regime: across eight controlled comparisons (deep Cox, GBT survival, multimodal methylation/mutation/CNA integration, engineered features), no alternative exceeded it, and several were significantly worse. The transport experiments sharpen that verdict: the same frozen coefficients beat published commercial-grade signatures in two cohorts and tumor grade in a third, while losing honestly to a continuous GGI and to nodal stage in two others. External replication, not internal cross-validation, is the honest arbiter of a prognostic claim; five of our nine discovery rounds survive only as negatives, and they are reported here alongside the positive transport results by design. Limitations: the panel is expression-only, cohorts differ in treatment era and endpoint definition, and the GSE20685 result suggests clinical covariates still carry signal the panel misses.

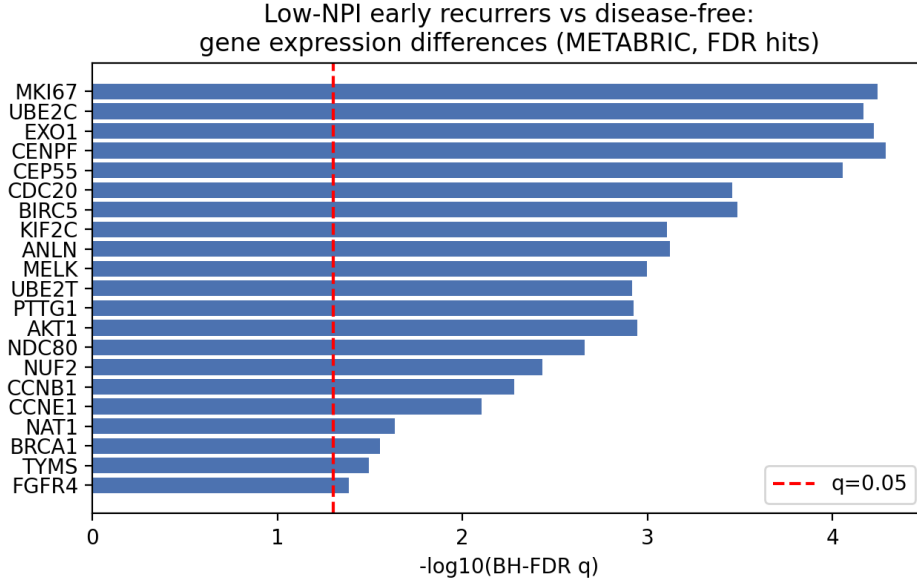


Figure 3: FDR-significant genes in the low-NPI discordance contrast.

## 6 Reproducibility

All code (41 hermetic tests), cached API snapshots, result JSON files, and figures are in the project repository; every number in this paper is regenerable from `results/*.json`. The audit-corrected resource roster has at most 36 distinct science/data tools before direct-use scrutiny and does not meet the 40-tool gate. Its 3,797 patient/sample identifiers are nested records, not independent study datasets. The 120-dataset gate is conditional on direct record-level use and uniqueness; the source-study count is nine and does not meet it.

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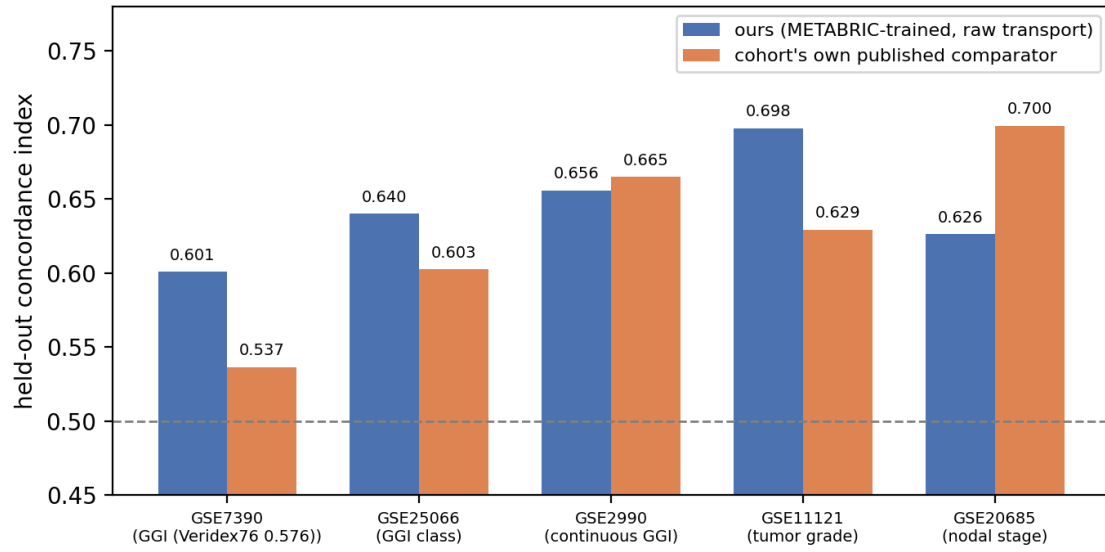


Figure 4: Transported model versus each cohort’s own published comparator across five independent cohorts; numbers from results/transport\*.json.

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